Adaptive basis sets for molecular orbitals

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1 Theory

1.1 Problem formulation

Question: What is the relation between N_e and N_{MO}?

Let us consider a system of N_e electrons. We solve the electronic structure problem

$$H\Psi = E\Psi \tag{1}$$

using post Hartree-Fock methods, such as Full Configuration Interaction (FCI) and Coupled Cluster (CC). The Hamiltonian under the second quantisation reads

$$\hat{H} = \sum_{\sigma} \sum_{ij}^{N_{\text{MO}}} h_{ij} \hat{a}_{i\sigma}^{\dagger} \hat{a}_{j\sigma} + \sum_{ijkl}^{N_{\text{MO}}} h_{ijkl} \hat{a}_{i\sigma}^{\dagger} \hat{a}_{k\sigma}^{\dagger} \hat{a}_{j\sigma} \hat{a}_{l\sigma}, \tag{2}$$

where N_{MO} is the number of basis functions used to discretize the molecular orbitals. The main idea is to first run a standard Hartree-Fock calculation using a discretization of the molecular orbitals based on N_b given atomic orbitals, yielding the coefficient matrix $C \in \mathbb{R}^{N_e \times N_b}$ in the decomposition

$$\forall 1 \leq i \leq N_e, \quad \phi_i = \sum_{\mu=1}^{N_b} C_{i\mu} \chi_{\mu}.$$

Then problem of eq. (1) is solved by expanding the wave function Ψ on a basis of N_{MO} molecular orbitals, as

$$\Psi = \sum_{i=1}^{N_{\text{MO}}} a_i \phi_i.$$

The coefficients $(a_i)_{1 \leq i \leq N_{\text{MO}}}$ are defined based on quantities related to the Hamiltonian of eq. (2).

1.2 Overview of methods

1.2.1 Existing work

Density fitting Density fitting has been used in order to compress the molecular orbital integrals of the post-Hartree-Fock equations (DFT correction method) using empirical auxiliary basis sets [1]. The difference of this applications when compared to conventional density fitting applications is that one compresses the molecular orbitals instead of the atomic orbitals (?).

1.2.2 Present work

Adaptive basis set The tensor h_{ijkl} is expensive to store in a quantum simulator, in terms of memory requirements. In order to solve this issue, we propose a new method that allows to reduce the number of basis functions used to discretize molecular orbitals, from a given atomic orbital basis set and a molecular geometry.

Tensor compression The main idea of this method is to compress the four-index tensor h_{ijkl}, where i, j, k, l are indices of molecular orbitals. The compression is based on the fact that this tensor can be precomputed as it is independent of the density matrix, which is updated during the iterative procedure. Therefore we can perform tensor compression entirely offline. This compression can be interpreted as a density fitting method performed on an auxiliary basis set that is generated on-the-fly. The difference of this method when compared to existing density fitting methods [1] is that the empirical auxiliary basis set of conventional density fitting is replaced by an auxiliary basis set generated on-the-fly based on the pivoted Cholesky decomposition.

1.3 Algorithms

1.3.1 Adaptive basis set generation

Our first attempt consists in proposing methods that reduce the size of a given atomic orbital basis set. Let $\{\chi_\mu\}_{1\leq\mu\leq N_b}$ denote a given atomic orbital basis set, with fixed atomic centers on the atomic positions of a given molecule. Note that the interatomic distances are fixed in this molecule. The first step is to assembly the Gram matrix (i.e. the overlap matrix) of this basis set under a user-defined inner product $\langle\cdot,\cdot\rangle$, denoted by $\boldsymbol{G}$ with entries defined as

$$1 \leq \mu, \nu \leq N_b, \quad G_{\mu\nu} = \langle\chi_\mu, \chi_\nu\rangle.$$

The pivoted Cholesky decomposition under a given tolerance value $\varepsilon > 0$, performed on the matrix $\boldsymbol{G}$, yields an approximate rank $M \leq N_b$ of $\boldsymbol{G}$ and a selection of M rows (and columns) that approximately span the full row space under the tolerance value. We propose to use the atomic orbitals corresponding to the selected indices to build the set of adaptive basis functions. This strategy leads to our algorithm named ABS-0, for Adaptive Basis Set generation.

Additionally, it may be of interest to impose maximal orbital type constraints to the adaptive basis set, in favour of computational efficiency within the integral evaluation procedure. The preliminary treatment of the Gram matrix $\boldsymbol{G}$ that enables this constraint can be presented as follows. The main idea is to construct an intermediary atomic orbital basis respecting orbital type constraints. We propose two possible methods. For each $1 \leq \mu \leq N_b$, if the atomic orbital χ_μ has orbital type strictly higher than d-type, then

(a) ABS-1: discard χ_μ.

(b) ABS-2: for each orbital type $t \in \{s, p, d\}$, create a new atomic orbital having the orbital exponent and atomic center of χ_μ, but orbital type t.

It is assumed that all atomic basis functions are normalized, so that the diagonal of $\boldsymbol{G}$ consists of one entries.

2 Numerical experiments

The first experiment allows to compare the results obtained from a DFT/B3LYP calculation on the water molecule at equilibrium geometry, using state-of-the-art auxiliary basis sets (RI and JKFIT) and our adaptive basis sets, obtained with methods ABS-x for $x = 0, 1, 2$. The aim of this comparison is to study which method achieves better accuracy with less basis functions. The approximation error is measured on ground state energies using the full atomic orbital basis set as a reference calculation. We are also interested in assessing the convergence of the approximation error of our method for varying tolerance values in the Cholesky decomposition.

References

[1] Andreas Hesselmann, Emmanuel Giner, Peter Reinhardt, Peter J Knowles, Hans-Joachim Werner, and Julien Toulouse. A density-fitting implementation of the density-based basis-set correction method. *arXiv preprint arXiv:2309.10390*, 2023.